

Revision Session Math 1

Wk 1 numbers. / straight line
set / Equation.
relation
fn.

$$\sqrt{2} \quad \sqrt{3} = p/q$$

$$\sqrt{3}$$

Note $\rightarrow \mathbb{N}_0 = \{0, 1, 2, \dots\}$

$$\text{ratio} = p/q \quad (p, q) = 1$$

$$\text{ratio} = \frac{p}{q} \quad \text{iff.} \quad \frac{p}{q} \quad q \neq 0$$

$$\sqrt{2}, \sqrt{3}$$

$$1.4142135611 \dots \quad p/q$$

$$\log 2 = p/q$$

$$-24$$

$$\frac{-24}{1} = p/q$$

real ration $\cup$ irr
 $\mathbb{Q} \quad \mathbb{Q}$

Set what

$$S = \{1, 2, 3, \dots, \text{Mistake} \dots \text{ten}\}$$

$$S = \{1, 4, 9, \dots\} \times \dots$$

$$S = \{n^2 \mid n \in \mathbb{N}\} = \text{Set}$$

$$\text{Set operation} = \underline{\underline{A \cup B}} = \{x \mid x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

$$A = \{1\}, B = \{1, 2\}$$

$$A \cup B = \{1, 2\} = B$$

$$A \cap B = \{1\} = A$$

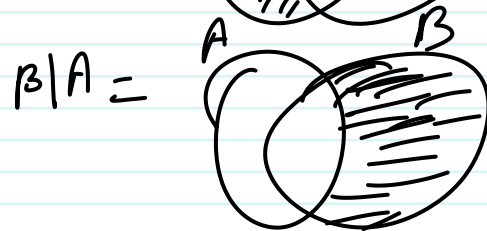
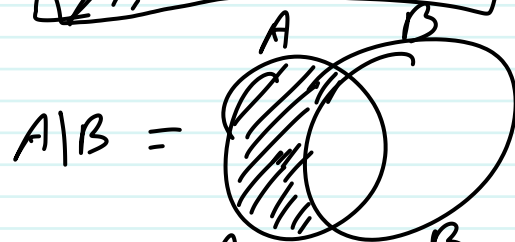
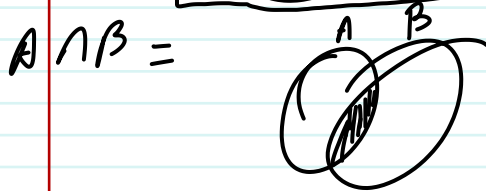
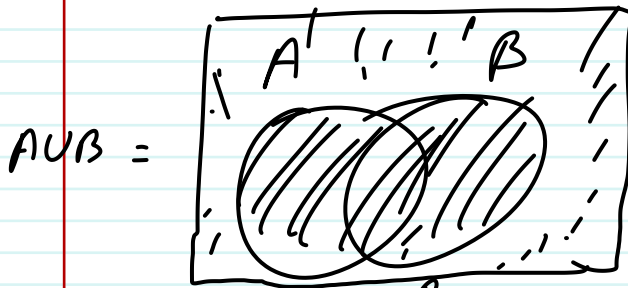
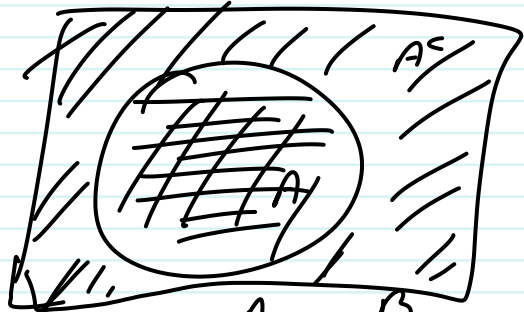
$$A \setminus B = \emptyset$$

$$B \setminus A = \{2\}$$

Complement $A = \{1\}$

$$A^c = \{2\}$$

super $B = \{1, 2\}$

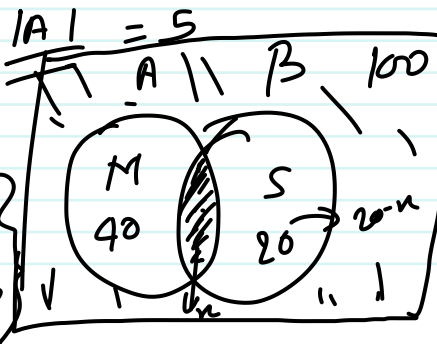


Cardinality of a set = Number of element A

$$|A| =$$

$$S = \{n^2 \mid n \in \mathbb{N}\}$$

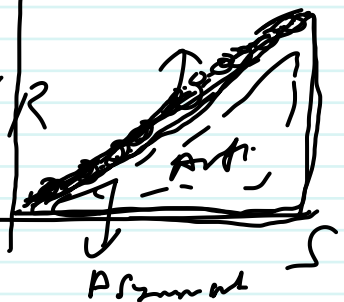
$$A = \{\underline{p}, \underline{q}, \underline{r}, \underline{s}, \underline{t}\}$$



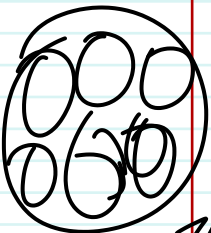
$$R \subseteq S \times S$$

$$(a, b), (b, a) \notin R$$

$$(a, b) \& (b, a) = (a, b)$$



$R \rightarrow$



$\mathbb{Z}$

$$\{(a, b) \mid \begin{matrix} a \bmod 2 = 0 \\ b \bmod 2 = 0 \end{matrix}\}$$

$$(1, 1) \quad (1, 1)$$

$$S = \{ \text{collide} \}$$

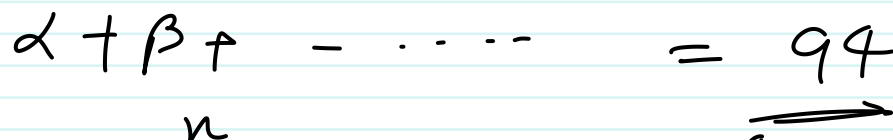


$$(y_i - \text{mean} - c)^2$$

$$(y_i - f(x_i))^2$$

$$x_i \in$$

$$\{x_1, x_2, \dots, x_n\}$$



A hand-drawn Venn diagram with four overlapping circles labeled M, F, S, and 4. An arrow points from M to F, and another arrow points from the intersection of F and S to the right.

$$S \times S = \{ (1,1) (1,2) (1,3), (2,1) (2,2) (2,3) (3,1) (3,2) (3,3) \}$$

$$= \{ (x,y) \mid x \in A, y \in B \}$$

$$\underline{A \times B} = \{(x, y) \mid x \in A, y \in B\}$$

$$\underline{R_2} = \left\{ \overset{2R_2}{(2,3)}, \overset{3R_2}{(3,3)} \right\} \text{ over } S$$

$$R = \left\{ \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \right\} \subseteq S \times S$$

S = collection fini

R on S $R = \{ \underset{\substack{= \\ xRy =}}{(x,y)} \mid x, y \in S \text{ and } x \text{ is your} \\ \text{bro } y \}$

Sym ref anti transi

$$S = \{a, b, c, d\}$$

Reflexive S and R is a relation S. $S \times S$

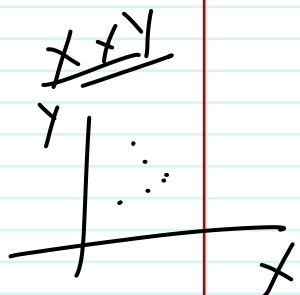
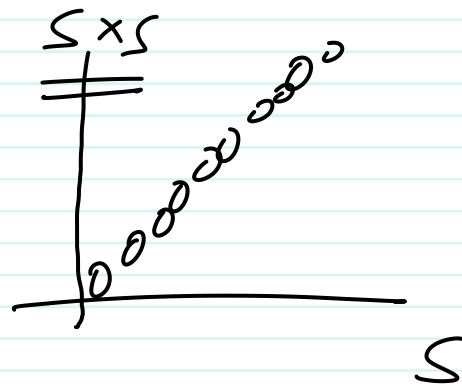
$$R = \{ \underline{(a,a)}, \underline{(b,b)}, \underline{(c,c)} \} \setminus \underline{(d,d)}$$

not Reflexive

Symmet R, S -

$$\begin{matrix} A & R & B \\ \downarrow & & \downarrow \end{matrix}$$

$$R \subseteq \underline{A \times B} \quad \underline{(a,b)} \in R \quad \underline{(b,a)} \in S$$



$$(1,2) \in X \times Y$$

$$\underline{A \times B}$$

$$\{1,2,3\}$$

$$\{a,b,c\}$$

$$(3,3)$$

$$A \times B = \{ \underline{(1,a)}, \underline{(1,b)}, \underline{(1,c)},$$

$$\underline{(3,a)}, \underline{(3,b)}, \underline{(3,c)} \}$$

$$R \subseteq A \times B = \{ \underline{(1,a)}, \underline{(3,b)} \}$$

$$A R B$$

$$(a,1)$$

$$A = \{1,2,\dots,3\}$$

$$B = \{0,1,\dots\}$$

$$\Rightarrow 1=2 \Rightarrow (a,c) \in R$$

$$R = \{ \underline{(1,2)}, \underline{(2,1)} \}$$

not reflexive

$$S = \{1,2,3\}$$

$$\underline{(1,2)}, \underline{(2,1)} \quad \text{Symmet}$$

(1,2) not transitive

$$R = \{ \underline{(1,2)} \} \quad (1,2), (2,1)$$

Anti

Anti Symmet

$$(1,2) \quad (2, \dots)$$

$$\underline{(a,b)} \in R, \Delta \underline{(b,a)} \in R \Rightarrow a=b$$

Anti

$$S = \{a, b, c\}$$

$$(a=a)$$

$$\{ \underline{(a,a)} \}$$

$$\underline{(a=a)}$$

$$\text{Sym } \{ \underline{(a,a)}, \underline{(b,b)}, \underline{(c,c)} \} \rightarrow \text{trans}$$

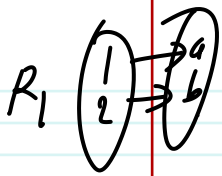
$$S = \{1,2,3\}$$

$$R = \{ \underline{(1,1)}, \underline{(2,2)} \}$$

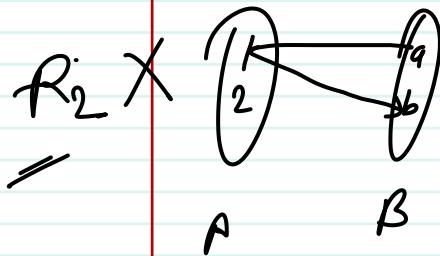
fn

$R \begin{matrix} \{1,2\} & \{a,b\} \\ \underline{A} & R & \underline{B} \\ \downarrow & & \downarrow \\ \text{domain} & & \text{co-domain} \end{matrix}$

$$A \times B = \{(1,a), (1,b), (2,a), (2,b)\}$$



$$R_1 = \{(1,a), (2,b)\} \subseteq A \times B$$

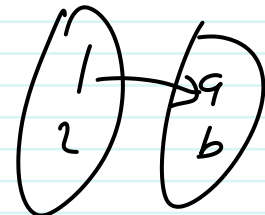


$$R_2 = \{(1,a), (2,a)\} \subseteq A \times B$$

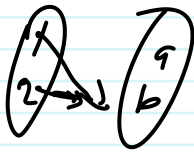
$$R_3 = \{(1,b), (2,b)\} \subseteq A \times B$$

$$R_4 = \{(1,a)\}$$

X R_4



R_3



R_4

$$f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^2 + 3x + 2 \quad x \in \mathbb{R}$$

$$R = \{(x, x^2 + 3x + 2) \mid x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}$$

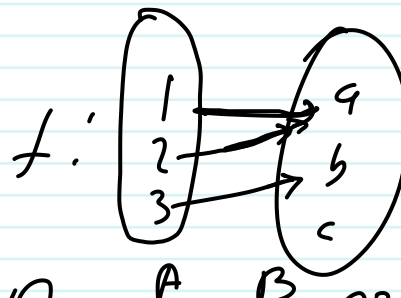
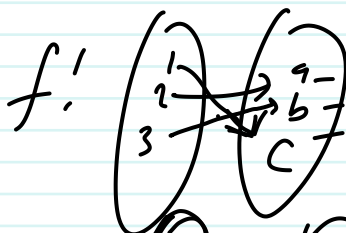
$$a \mapsto a^2 + 3a + 2$$

$$\underline{A \times B}$$

$$(x, y) \mid x \in B, y \in D$$

$$R = \{(x, y) \mid x^2 + y^2 = 1\} \subseteq \mathbb{R} \times \mathbb{R} \quad (y, x)$$

one-on and/or



R

$$R = \{(1,a), (2,a), (3,b)\}$$

not one-on

$$\text{Range} = \{a, b\}$$

not onto

co-domain of $\{a, b, c\}$

$$\text{Range} \neq \text{co-domain}$$

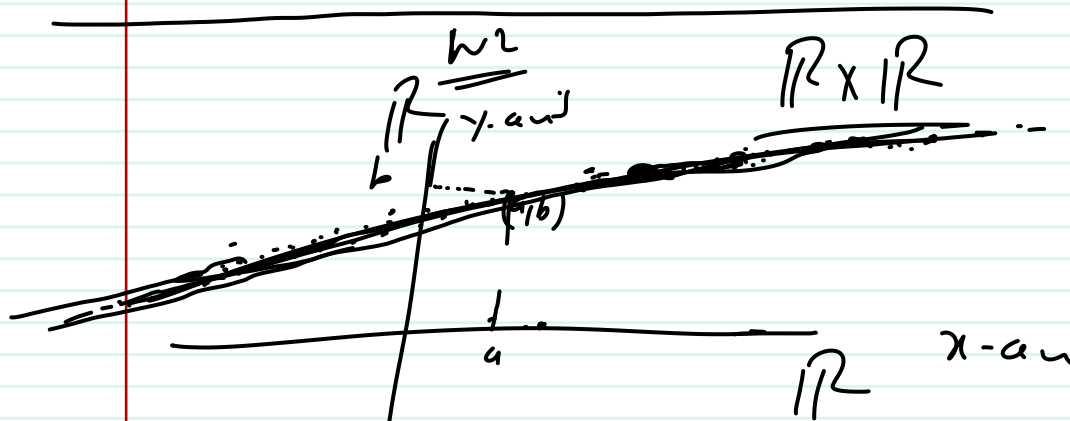
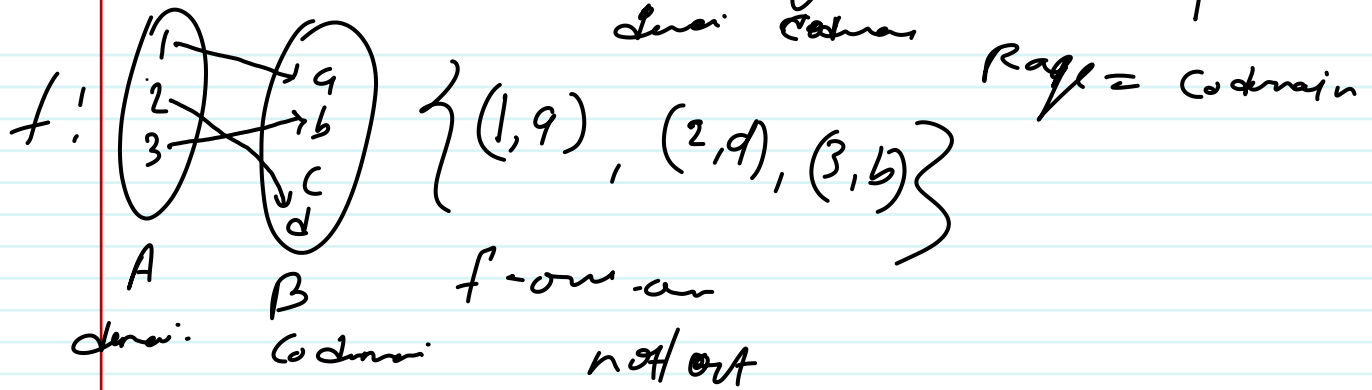
$$\{(1,c), (2,a), (3,b)\} \quad f: A \rightarrow B$$

$$\text{Range} = \{a, b, c\} = \text{co-domain}$$

onto

$$f: A \rightarrow B \quad \text{Range} = \{ f(a) \mid a \in A \}$$

$\downarrow$ $\downarrow$
 domain codomain



$$\{(a, b) \mid a, b \in \mathbb{R}\} = \mathbb{R} \times \mathbb{R}$$

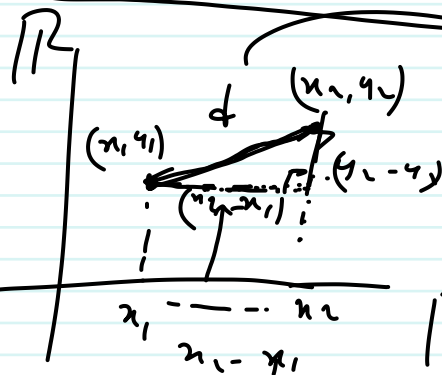
$$ax + by + c = 0$$

$$y = -\frac{a}{b}x - \frac{c}{b} \quad x = a.$$

$$2x + 6y + 8 = 0$$

$$4x + 12y + 16 = 0$$

$$2x + 6y + 8 = 0$$



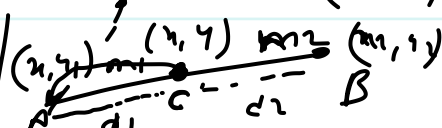
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\mathbb{R} \times \mathbb{R}$$

$$\mathbb{R} \quad A \quad B$$

$$(1, 2), (3, 4)$$

$$d = \sqrt{(3-1)^2 + (4-2)^2} =$$



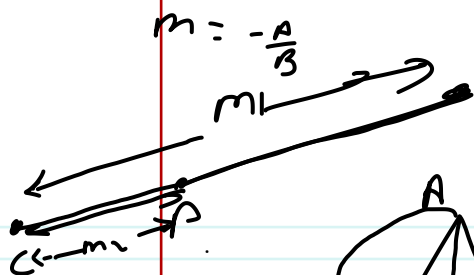
$$\frac{d_1}{d_2} = \frac{m_1}{m_2}$$

$$Ax + By + C = 0 \quad \text{Section}$$

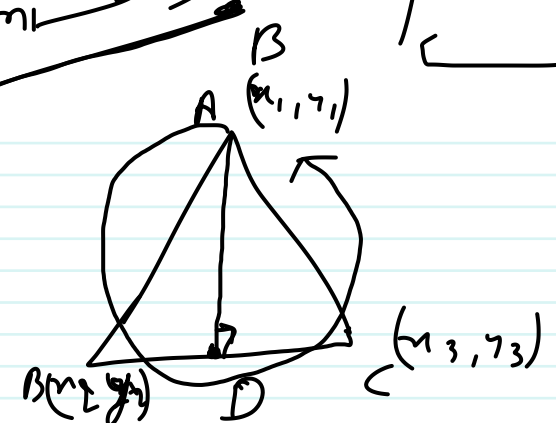
$$\Rightarrow By = -Ax - C$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$c = -\frac{C}{B}$$



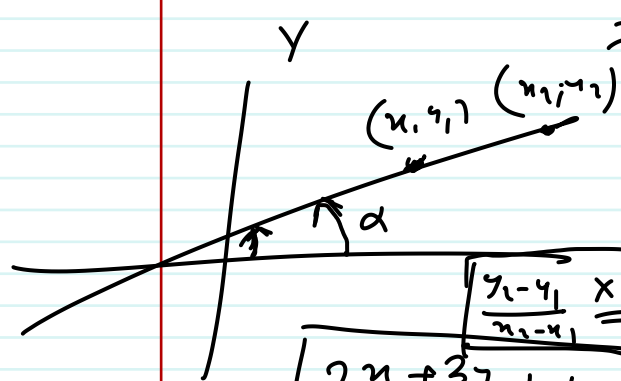
$$x = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$



$$\frac{1}{2} \left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right]$$

$\frac{1}{2} \times \text{height} \times \text{base}$

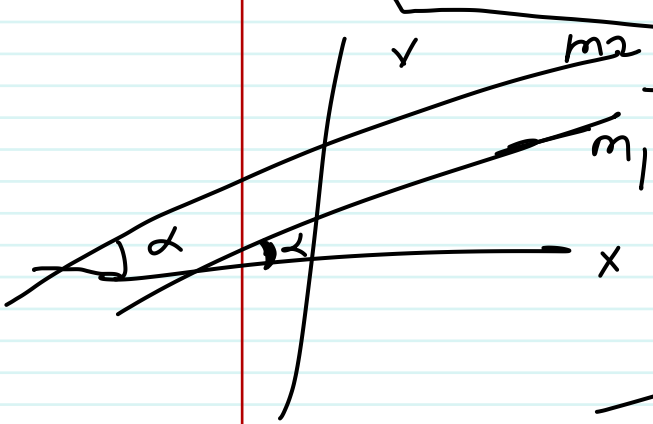
$$\frac{1}{2} \times AD \times BC$$



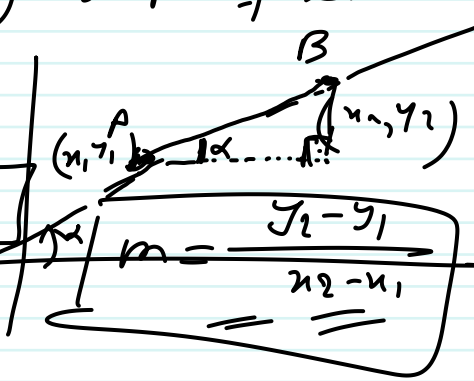
$$\frac{y_2 - y_1}{x_2 - x_1} = m = \tan \alpha$$

slope of line

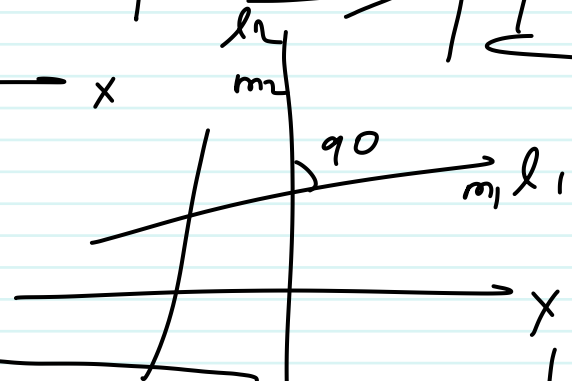
$$2x + 3y + 1 = 0$$



$$m_1 = m_2$$



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

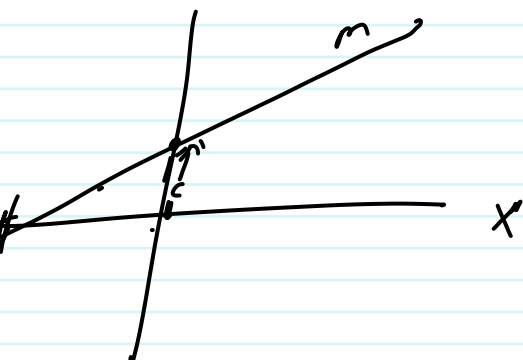


$$m_1 \cdot m_2 = -1$$

Slope int

$$y = mx + c$$

slope y-intercept



given $2x + 3y + 1 = 0$

$$3y = -2x - 1 \Rightarrow y = \left(-\frac{2}{3}\right)x - \left(\frac{1}{3}\right)$$

$$m = -2/3$$

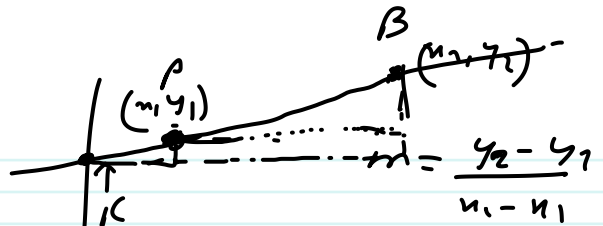
$$c = -1/3$$

$$y = (m)x + c$$

$$c = -1/3$$

$$ax + by + c = 0$$

$$y = \textcircled{m}x + \textcircled{c}$$



Two point form

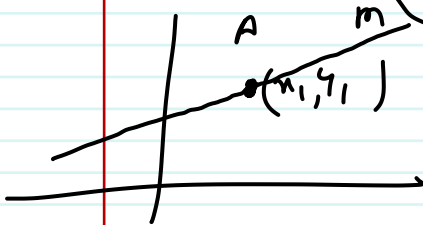
$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

$$\Rightarrow y = \frac{y_2 - y_1}{x_2 - x_1}$$

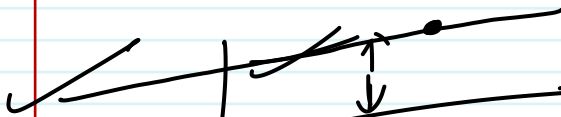
$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} x + \frac{y_2 - y_1}{x_2 - x_1} (-x_1)$$

$$y = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x + \left(y_1 - \frac{y_2 - y_1}{x_2 - x_1} x_1 \right) \Rightarrow c$$



$$\Rightarrow y - y_1 = m(x - x_1)$$



$$ax + by + c = 0$$

$$ax + by + c = 0$$

$$y = \left(\frac{-a}{b} \right) x - \frac{c}{b}$$

$$y - y_1 = -\frac{a}{b} (x - x_1)$$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$a_1x + b_1y + c_1 = 0$$

m_1

m_2

$|c_1 - c_2|$

$\sqrt{a_1^2 + b_1^2}$

$|c_1 - c_2|$

$\sqrt{a_1^2 + b_1^2}$

$|c_1 - c_2|$

$\sqrt{a_1^2 + b_1^2}$

$|c_1 - c_2|$

$\sqrt{a_1^2 + b_1^2}$

$|c_1 - c_2|$

$\sqrt{a_1^2 + b_1^2}$

$|c_1 - c_2|$

$$y - y_1 = \frac{b}{a} (x - x_1)$$

8

$$ax + by + c = 0$$

$$m_1 = -\frac{a}{b}$$

$$m_1 \cdot m_2 = -1$$

$$\frac{a}{b} \cdot m_2 = -1 \Rightarrow m_2 = -\frac{b}{a}$$